Supervised Admixture

2026-01-23

```
res_dir <- "../results/s05_b_supervised/"
```

This script uses supervised ADMIXTURE @alexanderFastModelbasedEstimation2009 to estimate ancestry proportions in Malagasy populations, using Borneo and Mozambique as reference populations.

Load gt

```
mad_bor_g1000_sum <-
  gt_load("../data/intermediate/rbind_data/mad_bor_g1000_sum_neo_clean_ld.gt",
    reattach_to = "../data/intermediate/rbind_data/mad_bor_g1000_sum_neo_clean_ld"
  )
```

Loci and sample count

```
nrow(mad_bor_g1000_sum)
```

```
## [1] 2744
```

```
count_loci(mad_bor_g1000_sum)
```

```
## [1] 198196
```

Subset mad_bor_g1000_sum

```
moz_pop <- unique(grep("^Mozambique",
  mad_bor_g1000_sum$population,
  value = TRUE
))

pops <- c(
  "Borneo", moz_pop, "Betsimisaraka", "Merina", "Diego",
  "Sakalava_Tsimihety", "Betsileo", "Vezo", "Mikea", "Temoro"
)

mad_bor_g1000_sum_sub <- mad_bor_g1000_sum %>% filter(population %in% pops)

mad_bor_g1000_sum_sub <- mad_bor_g1000_sum_sub %>% group_by(population)
```

Write plink files

```
gt_as_plink(mad_bor_g1000_sum_sub,
  file = "../data/intermediate/admixture/admixture_supervised_subset",
  overwrite = TRUE
)
```

```
## [1] "../data/intermediate/admixture/admixture_supervised_subset.bed"
```

Write .pop file

```
mad_bor_g1000_sum_sub <- mad_bor_g1000_sum_sub %>%
  mutate(Region = "-")
mad_bor_g1000_sum_sub <- mad_bor_g1000_sum_sub %>%
  mutate(Region = case_when(
    population %in% c("Borneo", "Sumba", "Timor") ~ "Asian",
    population %in% moz_pop ~ "African",
    TRUE ~ "-"
  ))

write.table(mad_bor_g1000_sum_sub$Region,
  file = "../data/intermediate/admixture/admixture_supervised_subset.pop",
  row.names = FALSE, col.names = FALSE, quote = FALSE
)
```

Run admixture

```
cd ~/Git/malagasy_diversity/code
bash s05_admixture_supervised.sh
```

```
## ****                  ADMIXTURE Version 1.3.0                 ****
## ****                  Copyright 2008-2015                     ****
## ****           David Alexander, Suyash Shringarpure,          ****
## ****                John  Novembre, Ken Lange                 ****
## ****                                                          ****
## ****                 Please cite our paper!                   ****
## ****   Information at www.genetics.ucla.edu/software/admixture  ****
##
## Random seed: 43
## Point estimation method: Block relaxation algorithm
## Convergence acceleration algorithm: QuasiNewton, 3 secant conditions
## Point estimation will terminate when objective function delta < 0.0001
## Estimation of standard errors disabled; will compute point estimates only.
## Supervised analysis mode.  Examining .pop file...
## Size of G: 299x198196
## Performing five EM steps to prime main algorithm
```

```
## 1 (EM)   Elapsed: 1.743  Loglikelihood: -6.45408e+07 (delta): 4.13285e+07
## 2 (EM)   Elapsed: 1.425  Loglikelihood: -6.30412e+07 (delta): 1.49958e+06
## 3 (EM)   Elapsed: 1.421  Loglikelihood: -6.26987e+07 (delta): 342510
## 4 (EM)   Elapsed: 1.43   Loglikelihood: -6.26021e+07 (delta): 96639.9
## 5 (EM)   Elapsed: 1.754  Loglikelihood: -6.25698e+07 (delta): 32275.3
## Initial loglikelihood: -6.25698e+07
## Starting main algorithm
## 1 (QN/Block)     Elapsed: 2.996  Loglikelihood: -6.19368e+07 (delta): 633033
## 2 (QN/Block)     Elapsed: 3.135  Loglikelihood: -6.18794e+07 (delta): 57335.7
## 3 (QN/Block)     Elapsed: 5.745  Loglikelihood: -6.18528e+07 (delta): 26657.8
## 4 (QN/Block)     Elapsed: 5.091  Loglikelihood: -6.18204e+07 (delta): 32339
## 5 (QN/Block)     Elapsed: 5.264  Loglikelihood: -6.17831e+07 (delta): 37353.5
## 6 (QN/Block)     Elapsed: 5.27   Loglikelihood: -6.17736e+07 (delta): 9515.09
## 7 (QN/Block)     Elapsed: 5.326  Loglikelihood: -6.17716e+07 (delta): 1987.27
## 8 (QN/Block)     Elapsed: 4.429  Loglikelihood: -6.17714e+07 (delta): 196.273
## 9 (QN/Block)     Elapsed: 3.893  Loglikelihood: -6.17714e+07 (delta): 22.0504
## 10 (QN/Block)    Elapsed: 3.954  Loglikelihood: -6.17713e+07 (delta): 8.40227
## 11 (QN/Block)    Elapsed: 4.024  Loglikelihood: -6.17713e+07 (delta): 0.38901
## 12 (QN/Block)    Elapsed: 4.32   Loglikelihood: -6.17713e+07 (delta): 0.0119854
## 13 (QN/Block)    Elapsed: 4.006  Loglikelihood: -6.17713e+07 (delta): 0.000283487
## 14 (QN/Block)    Elapsed: 5.298  Loglikelihood: -6.17713e+07 (delta): 2.27988e-06
## Summary:
## Converged in 14 iterations (79.889 sec)
## Loglikelihood: -61771344.461139
## Fst divergences between estimated populations:
##  Pop0
## Pop0
## Pop1 0.140
## Writing output files.
```

Read output

```r
predicted_q <- read_q_files("../data/intermediate/admixture")
```

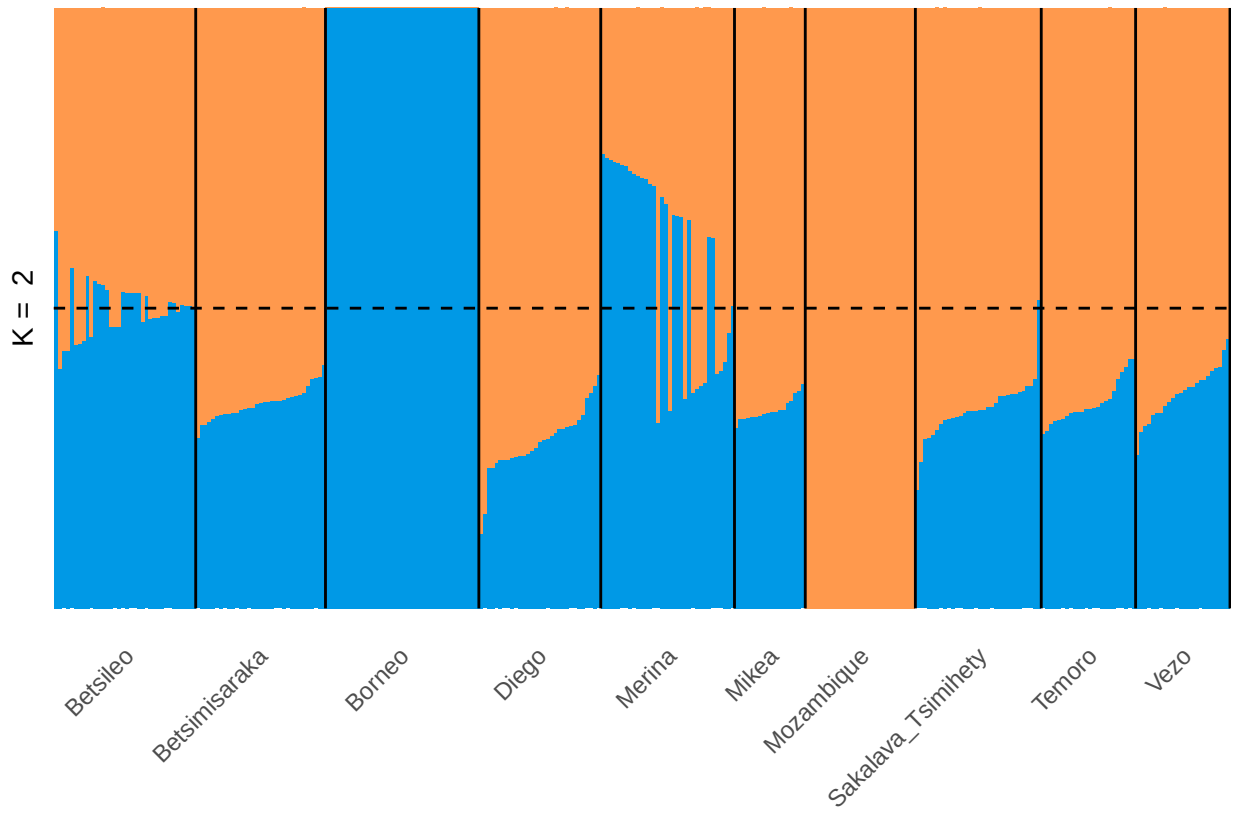
Plot

```r
# rename populations
mad_bor_g1000_sum_sub$population <-
  ifelse(mad_bor_g1000_sum_sub$population %in% moz_pop,
    "Mozambique", mad_bor_g1000_sum_sub$population
  )

# rename "Sakalava Tsimihety" and "Diego"
mad_bor_g1000_sum <- mad_bor_g1000_sum %>%
  mutate(population = case_when(
    population == "Sakalava_Tsimihety" ~ "Sakalava/Tsimihety",
    population == "Diego" ~ "Diego Mixed",
    TRUE ~ population
```

```
  ))

autoplot(get_q_matrix(predicted_q, k = 2, run = 1),
  data = mad_bor_g1000_sum_sub,
  annotate_group = TRUE, arrange_by_group = TRUE,
  arrange_by_indiv = TRUE, reorder_within_groups = TRUE
) +
  geom_hline(yintercept = 0.5, linetype = "dashed", color = "black")
```



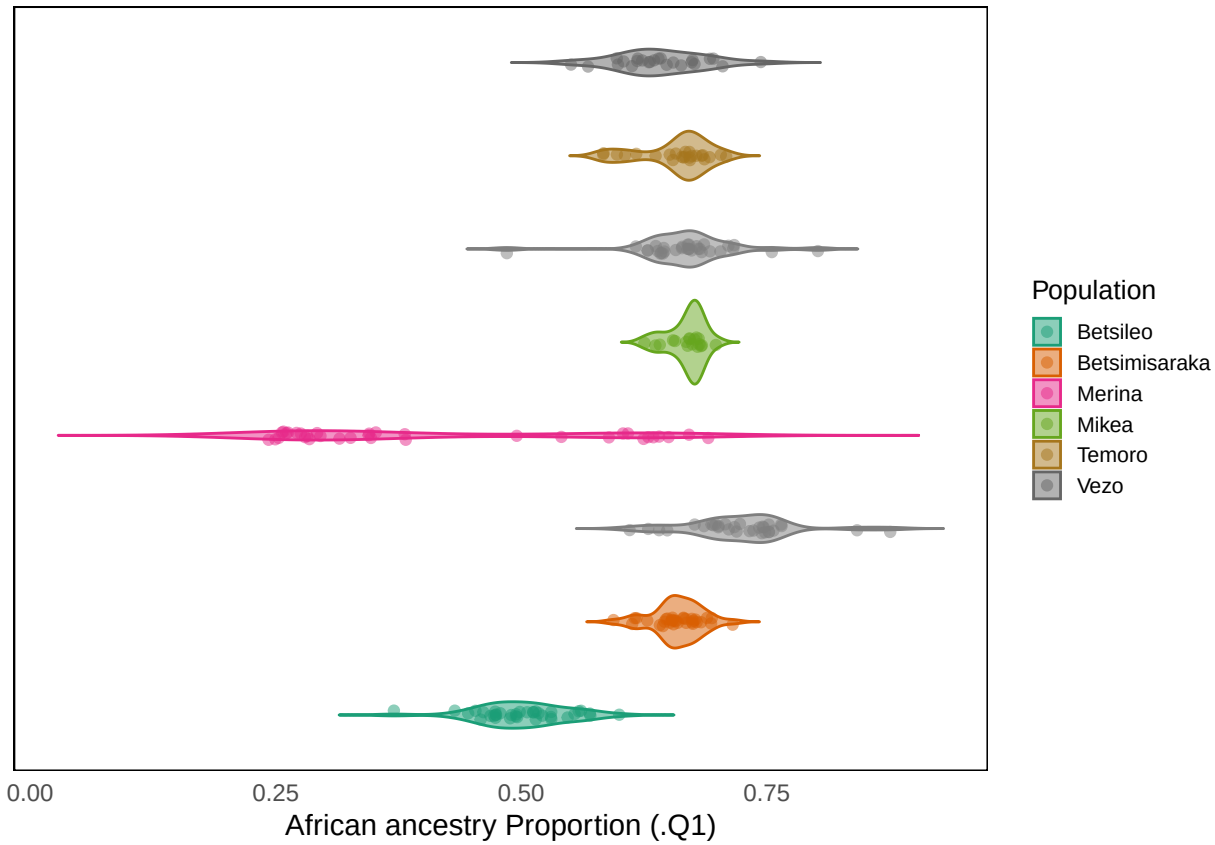

```
ggsave(paste0(res_dir, "Admixture_supervised_malagasy.jpg"),
  width = 7, height = 6, dpi = 300
)
```

```
q_tibble <- augment(
  get_q_matrix(predicted_q, k = 2, run = 1),
  mad_bor_g1000_sum_sub
)
q_tibble <- q_tibble %>% filter(!population %in% c("Borneo", "Mozambique"))

p1 <- ggplot(
  q_tibble,
  aes(
    x = .Q2, y = population,
    fill = population, colour = population
  )
```

```
) +
  geom_violin(alpha = 0.5, trim = FALSE) +
  labs(
    fill = "Population", colour = "Population",
    x = "African ancestry Proportion (.Q1)", y = NULL
  ) +
  geom_jitter(width = 0, height = 0.05, alpha = 0.5) +
  theme_minimal() +
  theme(
    axis.line = element_blank(),
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(),
    panel.background = element_rect(fill = "white"),
    legend.position.inside = c(0.15, 0.75),
    legend.text = element_text(size = 8),
    legend.title = element_text(size = 10),
    legend.key.size = unit(0.8, "lines"),
    axis.text.y = element_blank()
  ) +
  scale_colour_manual(
    values = c(
      "Betsileo" = "#1B9E77",
      "Betsimisaraka" = "#D95F02",
      "Diego Mixed" = "#7570B3",
      "Merina" = "#E7298A",
      "Mikea" = "#66A61E",
      "Sakalava/Tsimihety" = "#E6AB02",
      "Temoro" = "#A6761D",
      "Vezo" = "#666666"
    )
  ) +
  scale_fill_manual(
    values = c(
      "Betsileo" = "#1B9E77",
      "Betsimisaraka" = "#D95F02",
      "Diego Mixed" = "#7570B3",
      "Merina" = "#E7298A",
      "Mikea" = "#66A61E",
      "Sakalava/Tsimihety" = "#E6AB02",
      "Temoro" = "#A6761D",
      "Vezo" = "#666666"
    )
  )
p1
```

```
## Don't know how to automatically pick scale for object of type
## <q_matrix/q_matrix/matrix/array>. Defaulting to continuous.
```




```
ggsave(paste0(res_dir, "Afr_anc_violin_all_pops.jpg"),
  width = 7, height = 6, dpi = 300
)
```

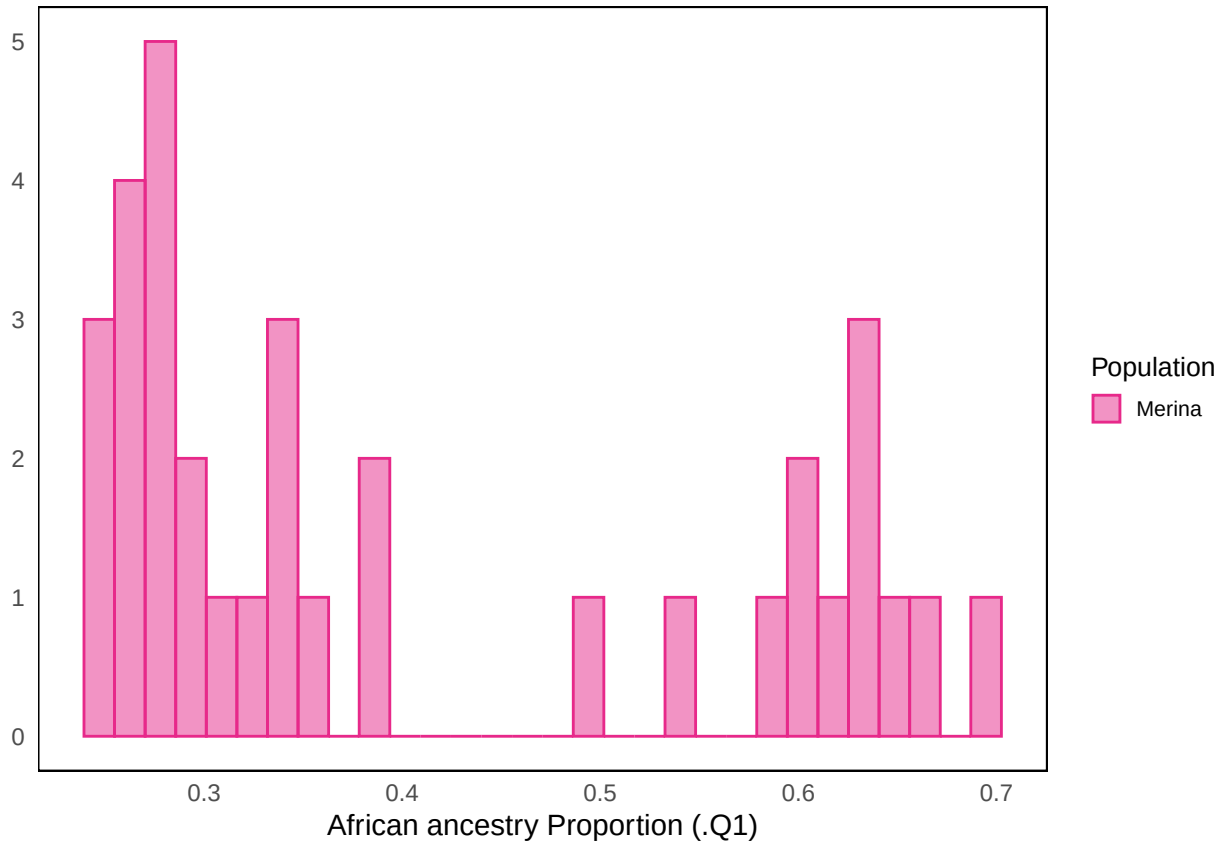
```
## Don't know how to automatically pick scale for object of type
## <q_matrix/q_matrix/matrix/array>. Defaulting to continuous.
```

```
q_tibble_merina <- q_tibble %>% filter(population == "Merina")

p2 <- ggplot(
  q_tibble_merina,
  aes(
    x = .Q2,
    fill = population, colour = population
  )
) +
  geom_histogram(alpha = 0.5) +
  labs(
    fill = "Population", colour = "Population",
    x = "African ancestry Proportion (.Q1)", y = NULL
  ) +
  theme_minimal() +
  scale_colour_manual(
    values = c(
      "Merina" = "#E7298A"
```

```
    )
  ) +
  scale_fill_manual(
    values = c(
      "Merina" = "#E7298A"
    )
  ) +
  theme(
    axis.line = element_blank(),
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(),
    panel.background = element_rect(fill = "white"),
    legend.position.inside = c(0.9, 0.9),
    legend.text = element_text(size = 8),
    legend.title = element_text(size = 10),
    legend.key.size = unit(0.8, "lines")
  )
p2
```

```
## Don't know how to automatically pick scale for object of type
## <q_matrix/q_matrix/matrix/array>. Defaulting to continuous.
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



```
ggsave(paste0(res_dir, "Afr_anc_hist_merina.jpg"),
  width = 7, height = 6, dpi = 300
)
```

```
## Don't know how to automatically pick scale for object of type
## <q_matrix/q_matrix/matrix/array>. Defaulting to continuous.
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```

mclust to confirm optimum number of clusters

```
classes_optimum <- Mclust(
  data = q_tibble_merina$.Q2,
  modelNames = "V"
)

clust_classes <- clustCombi(object = classes_optimum, q_tibble_merina$.Q2)

optimum <- clustCombiOptim(clust_classes)
optimum$numClusters.combi
```

```
## [1] 2
```
